



HEALTHCARE DATA ANALYSIS PROJECT REPORT

1. INTRODUCTION

This project focuses on analyzing healthcare data to extract meaningful insights related to patient demographics, disease patterns, treatment costs, insurance coverage, and hospital performance.

The complete workflow includes data preprocessing using Python, database management using PostgreSQL, analytical querying using SQL, and visualization using Power BI.

The objective is to convert raw healthcare data into actionable insights that can support better decision-making in healthcare systems.

2. PROBLEM STATEMENT

The dataset initially contained several issues such as:

- Missing and null values
- Inconsistent formatting across columns
- Unstructured data not suitable for direct analysis
- Lack of a centralized querying system

Due to these limitations, direct analysis and visualization were not possible. A structured data pipeline was required to clean, store, and analyze the data effectively.

3. PYTHON WORK (DATA CLEANING)

Python (Pandas library) was used for data preprocessing and cleaning.

Work performed:

- Loaded raw dataset into Pandas DataFrame
- Identified and handled missing values (replaced with 0 or appropriate values)
- Checked and corrected data types (numerical and categorical fields)
- Removed inconsistencies and duplicates

- Standardized column formats for analysis
- Exported cleaned dataset into CSV format for PostgreSQL import

Outcome:

A clean, structured dataset suitable for relational database storage and analysis.

4. POSTGRESQL WORK (SQL DATABASE)

A relational database was created in PostgreSQL to store structured healthcare data.

Work performed:

- Created a dedicated database for the project
- Defined table schema with appropriate data types
- Imported cleaned CSV file into PostgreSQL (1200+ records)
- Verified successful data insertion and integrity checks

Outcome:

A structured relational database enabling efficient SQL-based analysis and querying.

5. SQL ANALYSIS

SQL queries were used to perform detailed analysis on the dataset.

Key analyses performed:

- Total number of patients in the dataset
- Average, minimum, and maximum treatment costs
- Disease-wise patient distribution
- Insurance type vs treatment cost analysis
- Readmission rate analysis
- Age group-based healthcare insights
- Hospital performance comparison

Outcome:

SQL enabled fast and structured extraction of insights from large healthcare data.

6. POWER BI DASHBOARD

An interactive Power BI dashboard was developed to visualize key healthcare insights.

Dashboard features:

- KPI cards (Total Patients, Average Cost, Maximum Cost)
- Disease distribution charts
- Insurance type analysis
- Age group and gender-based insights
- Cost trends visualization
- Interactive filters and slicers for dynamic exploration

Outcome:

The dashboard provides a clear and interactive view of healthcare trends for better decision-making.

7. KEY INSIGHTS

The analysis revealed several important findings:

- Certain diseases contribute significantly higher treatment costs
- Elderly patients show higher risk and increased healthcare expenses
- Insurance type strongly influences total treatment cost distribution
- Readmission rates vary based on disease severity and patient condition
- Healthcare utilization patterns differ across age groups and demographics

8. CONCLUSION

This project demonstrates a complete end-to-end data analytics pipeline:

- Data cleaning and preprocessing using Python
- Structured storage and management using PostgreSQL
- Data analysis using SQL queries

- Visualization and reporting using Power BI

The workflow successfully transforms raw healthcare data into meaningful insights that can support healthcare planning and decision-making.

9. TOOLS USED

- ✓ Python (Pandas, NumPy)
- ✓ PostgreSQL
- ✓ SQL
- ✓ Power BI
- ✓ Microsoft Excel